ECA1011 - MICROPROCESSORS AND MICROCONTROLLERS

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EXP-1: Analysis and development verification of 8-bit and 16-bit arthemetic operation by using assesmbly programming language.

Objective:

1.8-bit add and sub using indel 8-bit microprocessor.

2.16-bit add and sub using indel 8-bit microprocessor.

3.implementation and testing of 8-bit multiplication and division using 8-bit microprocessor.

4. implementation and testing of 16-bit multiplication and division using 8-bit microprocessor.

Components used:

8085-simulator (or) 8085 intel micro processor.

Procedure:

1. Open the 8085 Simulator on the computer.
2. Write the assembly language program for 8-bit addition and subtraction.
3. Write the assembly language program for 16-bit addition and subtraction.
4. Develop programs for 8-bit multiplication and division.
5. Develop programs for 16-bit multiplication and division.
6. Execute the programs using suitable input values.
7. Check the results and verify the output.

EXp 1:- subb-8bit

MVI C 00h

LDA 4200h

4200	4A
4201	5A
4202	7A

MOV B3

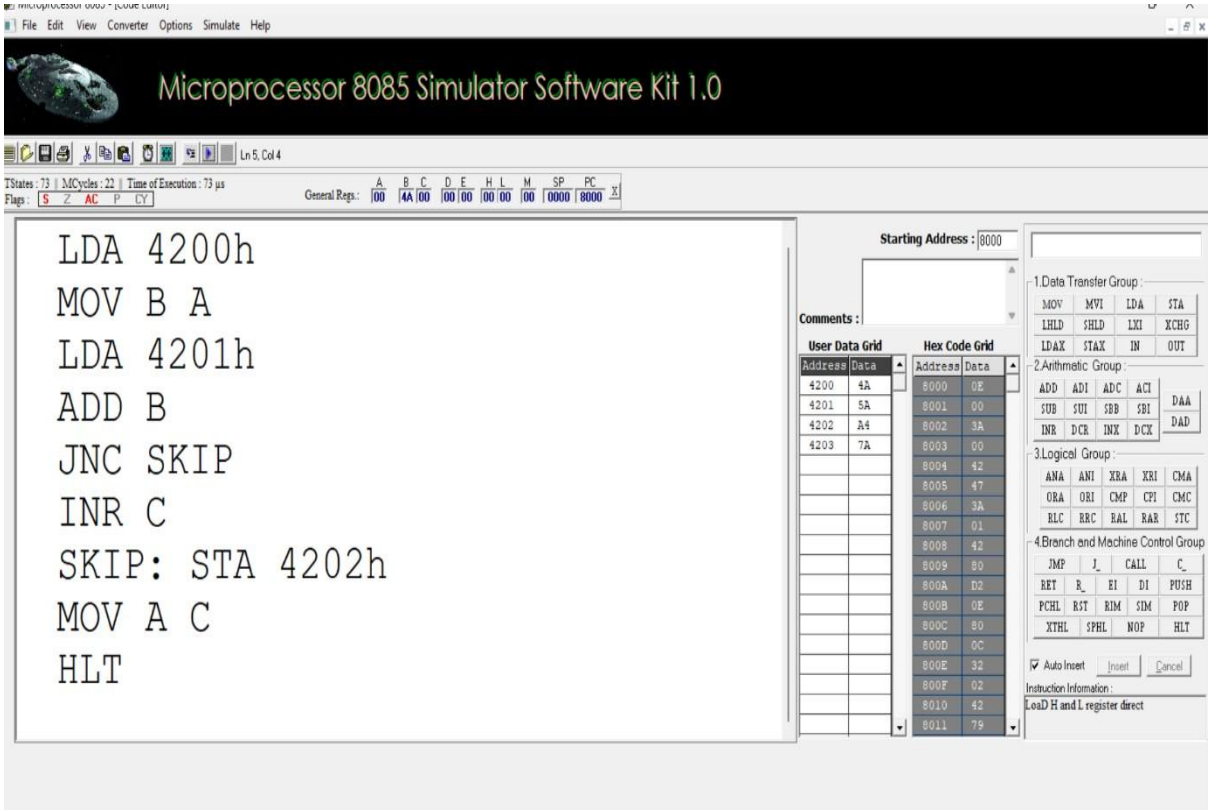
LDA 4201h 3

ADD JNC SKIP 3

INR3

SKIP: STA 4202h
MOV A3

HLT



EXP 1:-sub 16 bit

MVI C 00h

LDA 4201h

4200	25
4201	15
4202	00
4203	3A

MOV B A

LDA 4200h

SUB B

JNC SKIP

CMA

INR A

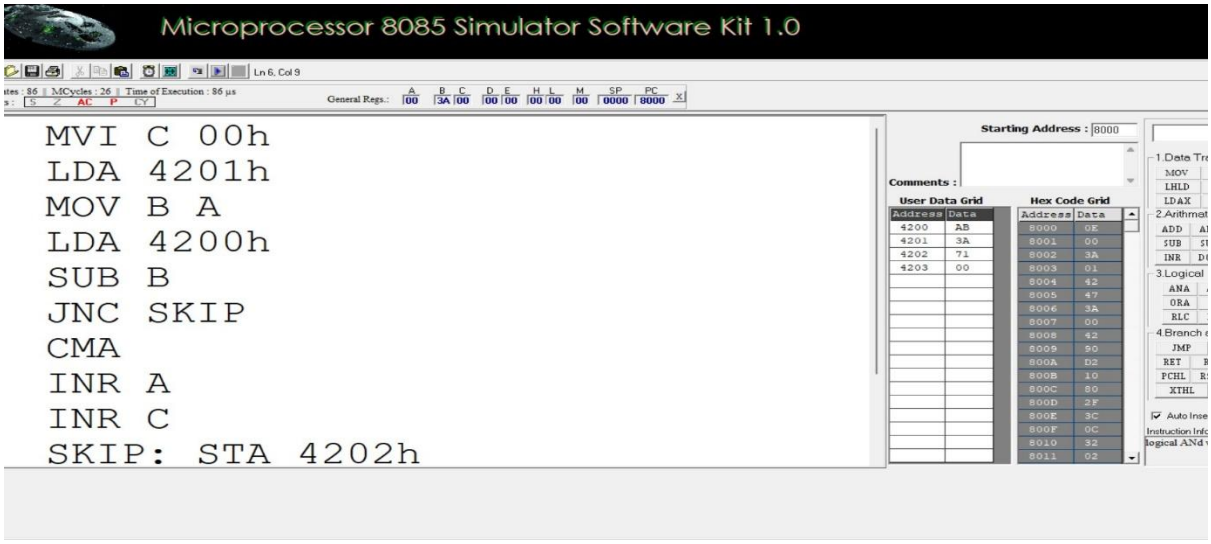
INR C

SKIP: STA 4202h

MOV A C

STA 4203h

HLT



16 bit addition

MVI C 00

LHLD 4000

XCHG

LHLD 4002

DAD D

JNC SKIP_CARRY

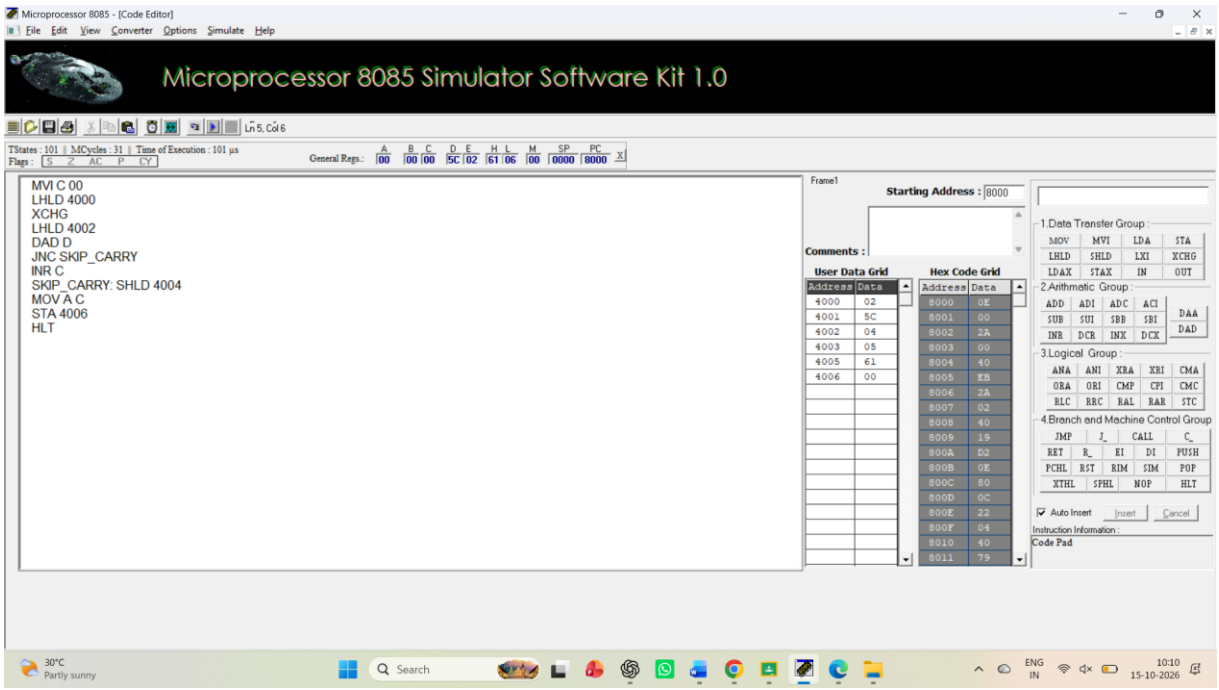
INR C

SKIP_CARRY: SHLD 4004

MOV A C

STA 4006

HLT



EXP-1.Multiplication and division of 8bit

LDA 4200h

MOV E A

LDA 4201h

4200	04
4201	05
4202	0C

MOV B A

LXI H 0000h

MVI D 00h

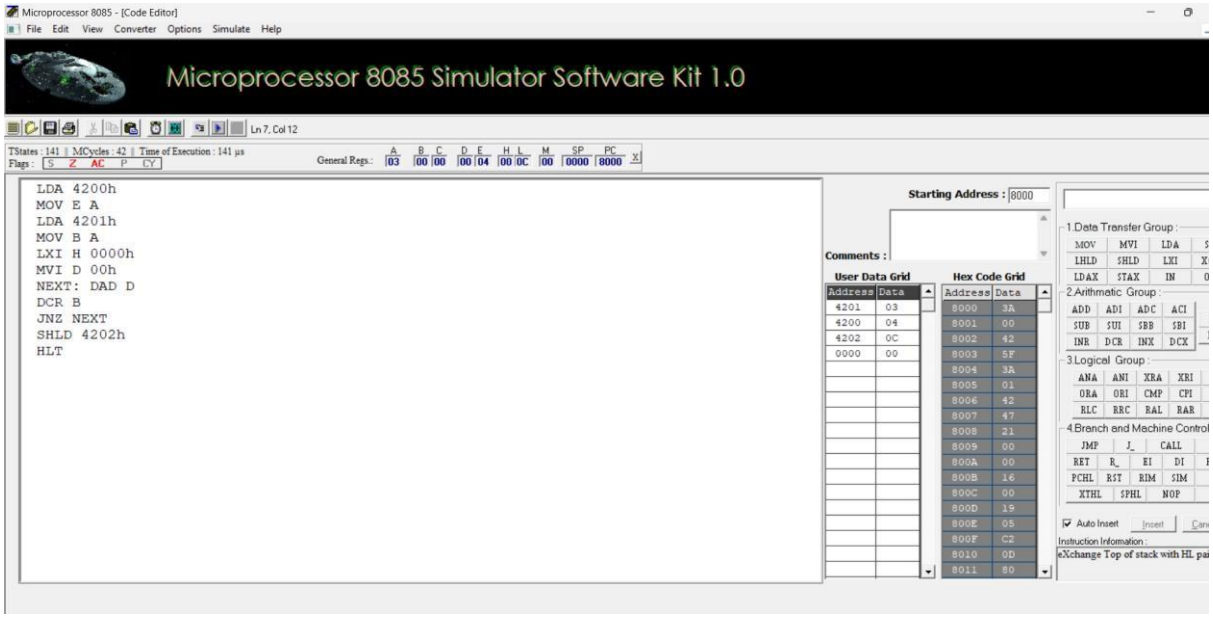
NEXT: DAD D

DCR B

JNZ NEXT

SHLD 4202h

HLT



EXP.1:-Division and multiplication of 16 bit

LDA 4201h

MOV B A

LDA 4200h

MVI C 00h

AGAIN: CMP B

4200	F5
4201	1A
4202	09
4203	0B

JC STORE

SUB B

INR C

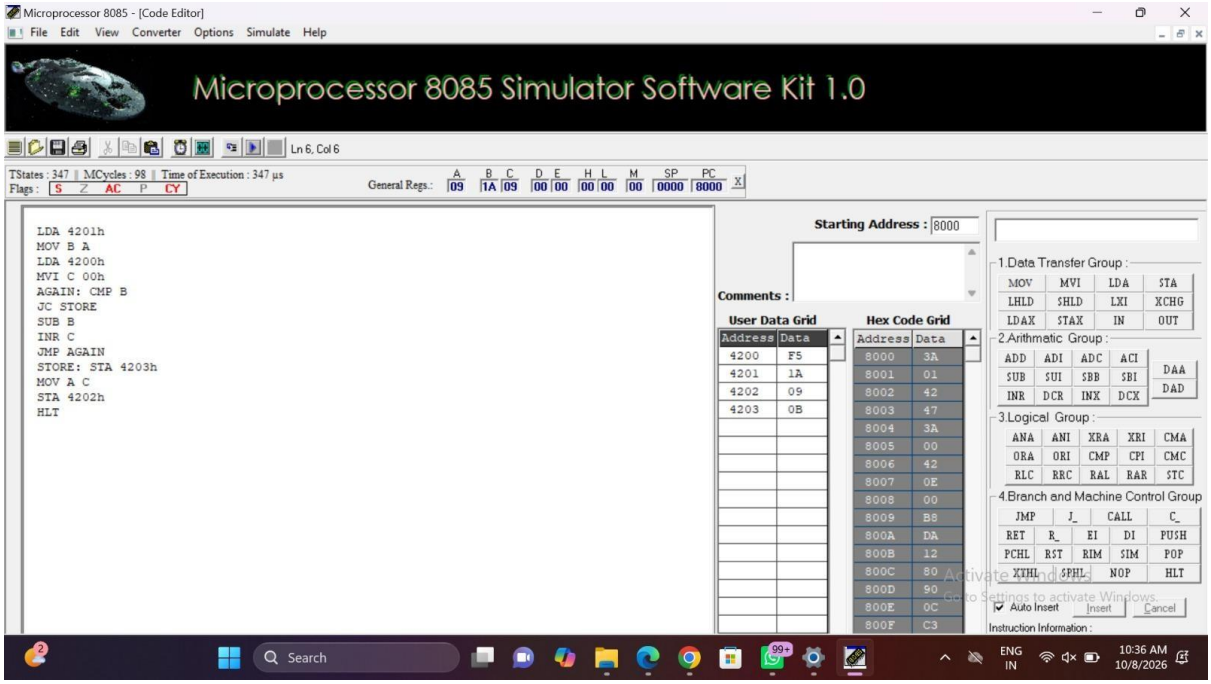
JMP AGAIN

STORE: STA 4203H

MOV A C

STA 4202h

HLT



multiplication of 16 bit

LXI H, 2050H

MOV E, M

INX H

MOV D, M

LXI H, 2052H

MOV C, M

INX H

MOV B, M

LXI H, 0000H

MVI A, 00H

LOOP: DAD D

DCR C

JNZ LOOP

SHLD 2054H

HLT

